Kelechi Emeka Ogbonna

Computational modelling & simulation · biology, autonomy and spaceflight systems

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PROFILE

Biotechnology graduate (M.Sc. in progress) who builds tested, reproducible simulation software: dynamical-systems models of disease and physiology, agent-based models of decision-making under uncertainty, and reinforcement-learning control. I separate measurements, simulations and hypotheses, report negative findings, and back claims with held-out evaluation and proper statistics. Seeking internships or graduate roles in computational modelling, space life sciences, or simulation and autonomy.

SELECTED PROJECTS

Adaptive Convergence: simulating R&D strategy for a spacefaring civilisation 2026

- Agent-based simulator of long-horizon technology development (procedural technology graphs with competing routes, hidden dead ends, Bayesian cost learning, recursive budget growth); six strategies benchmarked against a perfect-information oracle.
- Committing to the best option while abandoning projects on statistical evidence of overrun gave **+15.8% capability** over the best convergent baseline (95% CI 13.0–18.9%, better in 75% of 400 held-out worlds). It **cut budget lost to dead ends from 21% to 9%** and raised interstellar-milestone success from 41% to 56%.
- A 20-condition sweep showed the advantage exists only when failures lie outside the planner's model, and ablations showed extra exploration *reduced* performance. Found and fixed a belief-model bug, then reran all experiments. Replicated March (1991, *Organization Science*); held-out tuning seeds, bootstrap CIs, Holm-corrected Wilcoxon tests, 21 unit tests.

Python · NumPy · SciPy · pytest

Rocket Lander RL: reward shaping, reward hacking and robustness 2026

- Trained PPO agents to land a rocket (Gymnasium LunarLander) under four reward designs × 5 seeds, all evaluated on 500 landings with the unshaped reward. The trained agent landed on the pad **99.8%** of the time vs. **79.8%** for a hand-written control law.
- Reproduced reward hacking: a per-frame "upright and slow" bonus taught all 5 seeds to **hover until timeout (95% of episodes; 4.8% landings)**. Potential-based shaping (Ng et al., 1999) removed the hack and restored 98.2% landings.
- Monte Carlo dispersion tests: agents trained only in nominal conditions held 99% landings under heavier gravity (control law 49%), but all methods failed in maximum wind (≤ 21%).

Python · PyTorch · Stable-Baselines3 · Gymnasium · pytest

Project Confluence: shared metabolic dynamics across cancer types

- ODE state-space framework mapping six metabolic enzymes (HK2, PKM2, LDHA, IDH1/2, PDK1, G6PD) to model channels, with structural identifiability analysis and an adaptive controller.
- Validating against real CCLE metabolomics (225 metabolites × 928 cell lines) instead of synthetic data raised identifiable parameters from **7 to 15 of 17** across AML, osteosarcoma and NSCLC. About 43k lines of Python with **330+ tests** in GitHub Actions. Exposed to AI agents through an MCP server.

Python · SciPy · ODEs · identifiability · pytest · GitHub Actions · FastAPI · Model Context Protocol

Further modelling and software

- **Complexity Science**: gated pipeline that turns structured clinical case cards into ranked, evidence-tagged pathway hypotheses, without ever converting guideline text into model parameters (27 tests).
- **Individuality Dynamics**: coupled Kuramoto-oscillator experiment sweeping internal vs. external coupling, with multi-seed confidence intervals and Lempel–Ziv complexity.
- **Fly Brain vs. Tumor**: browser simulation in TypeScript/React in which a sparse mushroom-body circuit learns dosing on a toy cancer ODE (RK4) by reward-modulated plasticity.
- **TNBC metabolic strain model** (M.Sc. research): ODE and Monte Carlo model of ATP, ROS and glucose dynamics in triple-negative breast cancer, with stability and bifurcation analysis.
- **simplifyr** (Dev3Pack 2026 hackathon, Solana Mobile track): escrow app with a Rust/Anchor on-chain program, a Next.js/TypeScript front end and an LLM-based dispute arbiter.

RESEARCH WRITING

Independent computational thesis series 2026

About 50 computational manuscripts on identifiability, observability and control in disease and space-medicine models, indexed at research-theses-hub. Independent and not peer-reviewed; most include runnable Python. Selected:

- **Closed-loop agentic care planning for cervical cancer** (#52): care framed as a partially observed control problem with decision gates; synthetic worked example and study protocol. Zenodo, doi:10.5281/zenodo.23210715.
- **Identifiability of a shared metabolic cancer ODE** (#9): Fisher-rank analysis showing a lactate/glucose ratio identifies 1 of 5 rates and a multi-channel snapshot 4 of 5.
- **Space-medicine models**: microgravity bone loss on Mars-transit schedules (#45), radiation bystander effects under solar particle events (#48), and drug degradation on the lunar surface (#50).

EDUCATION

M.Sc. Biotechnology · Ahmadu Bello University, Zaria In progress

Systems biology, nonlinear dynamics, bioinformatics, bioprocess engineering.

B.Sc. Biotechnology (Hons) · Nile University of Nigeria, Abuja July 2022

Thesis: *In vitro antidiabetic activity of silver nanoparticles synthesised from Carica papaya leaf extract*: nanoparticle synthesis, characterisation, and α-amylase and antioxidant assays.

EXPERIENCE

Biotechnology Instructor · NYSC, Etinan College of Health and Technology, Akwa Ibom

- Designed and taught biotechnology modules, including introductory mathematical modelling of biological processes; led laboratory sessions and supervised student projects with limited equipment.

Research Intern · National Biotechnology Development Agency (NABDA), Abuja

- Molecular sequencing workflows and laboratory analysis on genetically modified crop projects; exposure to biosafety regulation.

Laboratory Intern · Arubah Medical Diagnostic Services, Calabar

- Processed clinical samples by PCR and spectrophotometry under laboratory quality-control procedures.

Managing Director · Uloma Shopping Mall | **POS Operator** · MoneyPoint Microfinance Bank

- Ran operations, finances, inventory and supplier relationships for a retail business; processed electronic payments in line with banking compliance procedures.

SKILLS

Programming	Python (NumPy, SciPy, Matplotlib, PyTorch, Stable-Baselines3, Gymnasium, FastAPI, pytest), TypeScript/React, MATLAB; working knowledge of Rust (Anchor); Git, GitHub Actions, Jupyter
Modelling	ODEs, stability and bifurcation analysis, structural and practical identifiability, Monte Carlo, agent-based simulation, Bayesian updating, reinforcement learning
Statistics	Bootstrap confidence intervals, paired and nonparametric tests, multiple-comparison correction, held-out evaluation design
Laboratory	PCR, sequencing workflows, spectrophotometry, nanoparticle synthesis, enzyme and antioxidant assays, BLAST

CERTIFICATIONS

IELTS General 7.5 · Google Project Management (Coursera) · Anthropic Academy: AI Fluency, Claude 101, Model Context Protocol (intro and advanced)